

A negative solution to MAIS-O87

Robert Sneiderman
Independent
robbysneiderman@gmail.com

September 2026

Abstract

In the coin-line model of MAIS-O87, the training coin sits at the right endpoint in most episodes, and the question is whether the learned preference for moving right can survive at a test state whose coin lies to the left. It cannot. For every fixed positive training diversity and every finite initialization, linear policy-gradient flow eventually assigns probability less than one half to moving right at the test state. The misgeneralization probability therefore tends to zero, which answers O87 in the negative. The correction time may depend on the diversity and on the initial parameters; no rate is claimed.

1 The question and the result

The agent moves on the sites $0, \dots, L$ and receives reward for reaching a coin. During training, the coin is usually at L , but with probability ε it is placed uniformly on $1, \dots, L$. A linear logistic policy chooses its direction from the agent's position and the coin's position. At the test state, the agent starts at $\lceil L/2 \rceil$ and the coin is at zero, so moving left is the correct action.

MAIS-O87 [2] asks whether, after arbitrarily long training, a positive fraction of Gaussian initializations can still favor moving right at this test state, uniformly over small positive ε . The following theorem rules this out. The coordinates (a, b, d) and the gradient flow are specified in the next section; $q_\varepsilon^{\text{RL}}(t; \sigma, L)$ denotes the probability that the test-state logit is positive at time t when $w(0) \sim N(0, \sigma^2 I_3)$.

Theorem 1. *Fix an integer $L \geq 4$ and a real $\varepsilon \in (0, 1]$. In the MAIS-A8 coin-line model, every finite initialization of linear policy-gradient flow satisfies*

$$a(t) + \lceil L/2 \rceil b(t) < 0 \quad \text{for all sufficiently large } t.$$

Consequently, for every $\sigma > 0$,

$$\lim_{t \rightarrow \infty} q_\varepsilon^{\text{RL}}(t; \sigma, L) = 0.$$

Thus the positive uniform lower bound asked for in O87 does not exist.

The conclusion is pointwise in ε . Each fixed positive value gives eventual correction, with no bound on the correction time. That is enough for O87, which takes the long-time limit with ε held fixed.

Idea of the proof. The parameter difference $A = d - b$ increases without bound, while every individual logit derivative tends to zero. At a state next to a coin, this forces the logit to tend to either $+\infty$ or $-\infty$. The main step is to rule out the possibility that all these adjacent states eventually favor moving right.

The reason is an asymmetry between the two sides of a coin. When the logits below the coin are positive, an episode that starts below and fails must make at least $\lceil (L+1)/2 \rceil$ left moves, each unlikely, so the gradient from that side decays like a high power of a small probability. Just above the coin, the adjacent state can refuse the coin by never stepping left, and the derivative there carries a single small factor, the probability of stepping left; every other factor is bounded below independently of the rest of the policy. The corrective gradient from above therefore decays more slowly than the gradient from below, and any positive training weight on coins left of L is enough to break the all-right pattern. Monotone bounds on the parameters then show that a nonnegative test-state logit at arbitrarily late times would force exactly that pattern. The proof does not need the training return to converge to one.

2 The model and its gradient

Write $w = (a, b, d)$ and $H = 2L$. At position p with coin at c , the logit and the probability of moving right are

$$z_{pc} = a + bp + dc, \quad r_{pc} = \frac{1}{1 + e^{-z_{pc}}}.$$

The position is $p \in \{0, \dots, L\}$; the training coin is $c \in \{1, \dots, L\}$, with probability

$$\mu_c = \frac{\varepsilon}{L} + (1 - \varepsilon)\mathbf{1}_{\{c=L\}} > 0.$$

Conditional on c , the initial position is uniform on $\{0, \dots, L\}$. An episode stops successfully on hitting c , including at time zero, and otherwise stops unsuccessfully after H steps. Left and right steps are clipped to $[0, L]$. The time-zero success and H -action conventions are those of MAIS-A8, Definition 3.1 [1]. The objective J is success probability, and $\dot{w} = \nabla J$ is Euclidean gradient ascent in these three coordinates.

Let J_c be conditional success probability. Temporarily regard all nonterminal logits for fixed c as independent variables, and define

$$y_{pc} = \text{sign}(c - p), \quad \kappa_{pc} = \left| \frac{\partial J_c}{\partial z_{pc}} \right|.$$

Increasing a logit helps below the coin and hurts above it. More precisely,

$$\frac{\partial J_c}{\partial z_{pc}} = y_{pc}\kappa_{pc}, \quad \kappa_{pc} > 0, \tag{1}$$

$$\nabla J = \sum_{c=1}^L \sum_{p \neq c} \mu_c \kappa_{pc} y_{pc} (1, p, c). \tag{2}$$

To prove (1), let $V_h(p, c)$ denote probability of hitting c within h steps, and let $\rho_t(p, c)$ be the probability, conditional on c , of occupying p at time t without having stopped. For fixed c , the Bellman recursion is

$$V_h(s, c) = (1 - r_{sc})V_{h-1}(s^-, c) + r_{sc}V_{h-1}(s^+, c) \quad (s \neq c),$$

with $V_0(s, c) = \mathbf{1}_{\{s=c\}}$ and $V_h(c, c) = 1$. Fix a nonterminal p and put $D_h(s) = \partial V_h(s, c)/\partial z_{pc}$. Differentiating one step yields

$$D_h(s) = (1 - r_{sc})D_{h-1}(s^-) + r_{sc}D_{h-1}(s^+) \\ + \mathbf{1}_{\{s=p\}}r_{pc}(1 - r_{pc})[V_{h-1}(p^+, c) - V_{h-1}(p^-, c)].$$

Here $D_0 = 0$ and $D_h(c) = 0$. Iterating this identity and averaging the initial state gives the occupancy weights ρ_t and hence

$$\frac{\partial J_c}{\partial z_{pc}} = r_{pc}(1 - r_{pc}) \sum_{t=0}^{H-1} \rho_t(p, c)[V_{H-t-1}(p^+, c) - V_{H-t-1}(p^-, c)], \quad (3)$$

where $p^+ = \min(p+1, L)$ and $p^- = \max(p-1, 0)$.

Below the coin, a path from p^- must hit p^+ before reaching c . The strong Markov property and monotonicity in the remaining horizon give $V_h(p^-, c) \leq V_h(p^+, c)$. Reflection gives the opposite inequality above the coin. At $h = H-1$ the inequality is strict: the closer neighbor has positive success probability, whereas from the farther neighbor there is positive probability of avoiding the closer one for all h steps. This uses self-loops at a wall or back-and-forth steps away from the closer neighbor. Since $\rho_0(p, c) = 1/(L+1)$, equation (3) has the strict claimed sign. Equation (2) is the chain rule.

3 Three monotone parameter combinations

The following combinations control how the parameters can diverge:

$$A = d - b, \quad R = A - a, \quad S = A + a.$$

The flow is globally defined. Indeed, J is a finite polynomial in finitely many logistic probabilities of linear forms. Its gradient and Hessian are globally bounded, so the gradient field is globally Lipschitz and bounded. Equation (2) gives

$$A' = \sum_{c, p \neq c} \mu_c \kappa_{pc} |c - p| > 0, \quad \|w'\| \leq \sqrt{1 + 2L^2} A'. \quad (4)$$

If A had a finite upper bound, (4) would imply finite total arclength and convergence of w to a finite point. At that point A' is strictly positive, a contradiction. Therefore

$$A(t) \longrightarrow +\infty. \quad (5)$$

Also $J' = \|\nabla J\|^2$ and $J \leq 1$. Thus the squared gradient has finite integral. It is uniformly continuous, since the gradient and Hessian are bounded. A nonnegative, uniformly continuous, integrable function tends to zero; hence $\nabla J \rightarrow 0$. Equation (4) implies $A' \rightarrow 0$ and, because every μ_c is positive,

$$\kappa_{pc}(t) \longrightarrow 0 \quad (p \neq c). \quad (6)$$

Finally,

$$R' = \sum_{c, p \neq c} \mu_c \kappa_{pc} (|c - p| - y_{pc}) \geq 0, \quad (7)$$

$$S' = \sum_{c, p \neq c} \mu_c \kappa_{pc} (|c - p| + y_{pc}) \geq 0. \quad (8)$$

Only the lower bounds $R \geq R(0)$ and $S \geq S(0)$, not divergence of R or S , are needed below.

4 The logits next to a coin

The two states next to each training coin carry the argument. Write

$$\ell_c = z_{c-1,c} \quad (1 \leq c \leq L), \quad q_c = z_{c+1,c} \quad (1 \leq c < L).$$

Lemma 1. *Each adjacent logit tends either to $+\infty$ or to $-\infty$.*

Proof. At a left-adjacent state, retain $t = 0$ in (3). Right immediately succeeds. After choosing left, success requires a later right action at $c - 1$. Choosing left at every subsequent visit to $c - 1$ for $H - 1$ steps prevents success, with probability at least $(1 - r)^{H-1}$. Therefore

$$\kappa_{c-1,c} \geq \frac{r(1-r)^H}{L+1}. \quad (9)$$

Similarly, at a right-adjacent state,

$$\kappa_{c+1,c} \geq \frac{r^H(1-r)}{L+1}. \quad (10)$$

These estimates are independent of all other policy probabilities. Precisely, modify the policy at the adjacent state to choose deterministically away from the coin. Under the modified policy, hitting the coin from the away neighbor is impossible. The original probability of these paths is the expectation of r^N , or $(1-r)^N$, with $N \leq H - 1$. This also includes clipped wall actions.

On a compact interval of logits the right sides of (9)–(10) have positive lower bounds. Thus (6) makes the absolute value of each adjacent logit tend to infinity. Continuity then gives a fixed eventual sign. $\square$

5 The adjacent logits cannot all tend to positive infinity

Lemma 2. *No trajectory has $\ell_c \rightarrow +\infty$ for every c and $q_c \rightarrow +\infty$ for every $c < L$.*

The lemma places no condition on logits farther from the coin, so the estimates below must hold whatever the policy does at those states.

5.1 Gradient estimates

Write $\nabla J_c = P_c - N_c$, where P_c is the contribution from $p < c$ and N_c is the magnitude of the contribution from $p > c$. Both are componentwise nonnegative by (1); $N_L = 0$. Let $F_c(p_0) = 1 - V_H(p_0, c)$. A path starting below the coin stops before visiting any state above it, so

$$P_c = \frac{1}{L+1} \sum_{p_0 < c} \nabla V_H(p_0, c) = -\frac{1}{L+1} \sum_{p_0 < c} \nabla F_c(p_0).$$

For the finite set $\mathcal{F}_c(p_0)$ of failed length- H action paths,

$$\nabla F_c(p_0) = \sum_{\omega \in \mathcal{F}_c(p_0)} \Pr_w(\omega \mid p_0, c) \nabla \log \Pr_w(\omega \mid p_0, c).$$

Put $n = \lceil (L+1)/2 \rceil \geq 3$. Constants $C, K > 0$ depending only on L satisfy the following estimates for every coordinate j . If $m_c = \min_{p < c} z_{pc} \geq 0$, then

$$(P_c)_j \leq C e^{-nm_c}. \quad (11)$$

For $c = 1$ and $h = z_{01} \geq 0$,

$$(P_1)_a = (P_1)_d \leq Ce^{-Hh}, \quad (P_1)_b = 0. \quad (12)$$

If $q_c \geq 0$ and $c < L$, then, without restrictions on other logits,

$$(N_c)_j \geq Ke^{-q_c}. \quad (13)$$

One may take $C = HL2^H$ and $K = 2^{-(H+1)}/(L+1)$.

For (11), a failed path starting below c stays below c . If it makes N_- left moves, its final position is at least $p_0 + H - 2N_-$; clipping at zero only increases this lower bound. Consequently $N_- \geq (H + p_0 - c + 1)/2$, and hence $N_- \geq n$. Each left probability is at most e^{-m_c} , so $\Pr_w(\omega \mid p_0, c) \leq e^{-nm_c}$ when $m_c \geq 0$. Each of the H factors of $\log \Pr_w(\omega \mid p_0, c)$ has partial derivatives bounded by L in absolute value, so the score is at most HL in every coordinate. There are at most 2^H action paths and at most L starting positions, each of weight $1/(L+1)$, and (11) follows with $C = HL2^H$. For $c = 1$, failure from below requires H left self-loops at $p = 0$, proving (12). For (13), use (10), $r \geq 1/2$, and $r(1-r) \geq e^{-q_c}/4$. All components of $(1, c+1, c)$ are at least one.

5.2 Proof of the exclusion lemma

Proof of Lemma 2. Suppose all adjacent logits tend to $+\infty$. At sufficiently late times $A > 0$ and $h = \ell_1 > 0$. Whenever $b \geq 0$, we have $d = A + b > 0$ and

$$m_c = h + (c-1)d, \quad q_1 = h + 2b.$$

Only $c \geq 2$ contribute positively to b' . With $\mu_1 = \varepsilon/L$,

$$b' \leq Ce^{-n(h+d)} - \mu_1 Ke^{-(h+2b)}. \quad (14)$$

The ratio of the positive term to the negative term is at most

$$\frac{C}{\mu_1 K} \exp(-(n-1)h - (n-2)b - nA),$$

which tends uniformly to zero as $A \rightarrow \infty$ for $h, b \geq 0$. Thus, at sufficiently late times, $b' < 0$ whenever $b \geq 0$. A first late crossing from $b < 0$ to $b \geq 0$ would occur at $b = 0$ with nonnegative left difference quotient, contradicting $b' < 0$ there. Hence no such crossing is possible.

Two cases remain. If b stays nonnegative, it is bounded and decreasing, while $d \rightarrow \infty$ and $h \rightarrow \infty$ by hypothesis. The positive a and d components are bounded by

$$Ce^{-Hh} + Ce^{-n(h+d)},$$

while their negative contribution from $c = 1$ is at least $\mu_1 Ke^{-(h+2b)}$. Since b is bounded, both positive-to-negative ratios tend to zero. Thus $a', d' < 0$ eventually, contradicting $h = a + d \rightarrow \infty$.

Otherwise $b < 0$ eventually. For $c < L$,

$$m_c = \ell_c = q_c - 2b \geq q_c.$$

As $q_c \rightarrow \infty$, equations (11) and (13) give, in every coordinate,

$$(\nabla J_c)_j \leq -\frac{K}{2}e^{-q_c} \quad (15)$$

eventually. For $c = L$, use the exact identity

$$m_L = \ell_L = q_{L-1} + A.$$

Its positive gradient is at most $Ce^{-n(q_{L-1}+A)}$, which is eventually smaller than half the weighted negative contribution of coin $L-1$ in (15). Every component of ∇J is therefore strictly negative eventually. This gives $q'_1 = a' + 2b' + d' < 0$, contradicting $q_1 \rightarrow \infty$. Both cases are impossible. $\square$

6 From the adjacent states to the test state

The last step connects the adjacent states to the test state. Let $k = \lceil L/2 \rceil \geq 2$ and let $P = a + kb$ be the test-state logit. Along any sequence of times tending to infinity with $P \geq 0$, every adjacent logit tends to $+\infty$; the two signs of b are treated separately. Write $R_0 = R(0)$ and $S_0 = S(0)$.

If $b \geq 0$, substituting $a = P - kb$ into $S = A + a \geq S_0$ gives

$$b \leq \frac{A + P - S_0}{k}.$$

Consequently

$$h = a + d = P + A - (k-1)b \geq \frac{A+P}{k} + \frac{k-1}{k}S_0 \geq \frac{A}{k} + \frac{k-1}{k}S_0. \quad (16)$$

For $A > 0$, $d = A + b > 0$, and every training logit satisfies $z_{pc} = h + bp + (c-1)d \geq h$. Equation (5) proves the claim in this case.

If $b < 0$, write $B = -b > 0$. Then $a = P + kB$, and $R = A - a \geq R_0$ gives

$$kB \leq A - P - R_0.$$

Writing $D = b + d = A - 2B$, we have

$$D = R + P + (k-2)B \geq R_0, \quad q_1 = P + A + (k-3)B.$$

For $k \geq 3$, $q_1 \geq A$. For $k = 2$,

$$q_1 \geq \frac{A}{2} + \frac{3P}{2} + \frac{R_0}{2} \geq \frac{A}{2} + \frac{R_0}{2}.$$

At sufficiently late times $A > 0$. In either case $q_1 \geq A/2 - |R_0|/2$. Since $q_c = q_1 + (c-1)D$,

$$q_c \geq \frac{A}{2} - L|R_0| \quad (1 \leq c < L). \quad (17)$$

The other adjacent logits satisfy $\ell_c = q_c + 2B$ for $c < L$ and $\ell_L = q_{L-1} + A$. They too diverge positively by (5). This completes the claim for both signs of b .

7 Proof of the theorem

Proof of Theorem 1. If $P(t) \geq 0$ at arbitrarily late times, the previous section supplies an unbounded sequence of times along which all adjacent logits are positive. Lemma 1 says that each has a fixed eventual sign and diverges in magnitude. All must therefore tend to $+\infty$, contradicting Lemma 2.

Thus $P(t) < 0$ for all sufficiently large t , from every finite initial vector. The indicator $\mathbf{1}_{\{P(t) > 0\}}$ tends pointwise to zero and is bounded by one. Bounded convergence under $N(0, \sigma^2 I_3)$ gives $q_\varepsilon^{\text{RL}}(t; \sigma, L) \rightarrow 0$. The same conclusion holds for any probability law on finite initial vectors. $\square$

For each fixed positive ε , the limsup in O87 is therefore zero. In particular, it cannot have a positive lower bound uniform over small positive ε .

Acknowledgments

Computations and draft review used Astra and Fable 5.1.

References

- [1] MAIS project (github.com/lionellevine/MAIS), *A predictive theory of out-of-distribution generalization: which of two perfect policies does gradient descent select?* Research agenda MAIS-A8, Definition 3.1 and Question 6.2, [pinned source](#), [commit 7b23a8a](#).
- [2] MAIS project (github.com/lionellevine/MAIS), *Does policy gradient stay misgeneralized despite training diversity?* Open problem MAIS-O87, [problem statement](#), [commit 7b23a8a](#).